

13/10/22

UNIT-11

Conformal mapping

Standard transformations - Bilinear transformation
 Taylor's & Laurent's series - Classification of singularities

Bilinear transformation (or) Linear fractional Transformation
 (or) mobius transform.

The transformation $w = \frac{az+b}{cz+d}$ where, $ad-bc \neq 0$ & a, b, c and d are complex constants. is called Bilinear transformation.

Define Fixed (or) Invariant points

The invariant points of the transformation are such that image of z itself

$$\text{i.e., } w = f(z) = z$$

Eg Questions:

1) Find the fixed points of the transformation given

$$w = \frac{1+z}{1-z}$$

Soln:

W.K.T

$$w = z$$

$$z = \frac{1+z}{1-z}$$

$$z - z^2 = 1 + z$$

$$z^2 = -1$$

$$z = \pm \sqrt{-1}$$

$$z = \pm i$$

$$z = i, -i$$

2. Find Fixed points for $W = \frac{2z-5}{z+4}$

Soln:

W.K.T

$$W = z$$

$$z = \frac{2z-5}{z+4}$$

$$z^2 + 4z = 2z - 5$$

$$z^2 + 2z + 5 = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{4 - 20}}{2}$$

$$= \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

$$\boxed{z = -1 \pm 2i}$$

6m (or) 11m (8)

Bilinear transformation:

Formula:

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

1) Find the Bilinear transformation that maps from
 $z = (1, i, -1)$ into $w = (2, i, -2)$

Soln:

$$z_1 = 1$$

$$w_1 = 2$$

$$z_2 = i$$

$$w_2 = i$$

$$z_3 = -1$$

$$w_3 = -2$$

W.K.T

$$\frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} = \frac{(\cancel{z} - \cancel{z}_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\frac{(w - 2)(i + 2)}{(2 - i)(-2 - w)} = \frac{(z - 1)(i + 1)}{(1 - i)(-1 - z)}$$

$$\frac{(w - 2)}{(-2 - w)} = \frac{(z - 1)(i + 1)(2 - i)}{(-1 - z)(1 - i)(i + 2)}$$

$$\frac{(w - 2)}{(-2 - w)} = \frac{(z - 1)(2i - i^2 + 2 - i)}{(-1 + z)(i + 2 - i^2 - 2i)}$$

$$\frac{(w - 2)}{(-2 - w)} = \frac{(z - 1)(i + 1 + 2)}{(-1 - z)(-i + 2 + 1)}$$

$$\frac{(w - 2)}{(-2 - w)} = \frac{(z - 1)(i + 3)}{(-1 - z)(-i + 3)}$$

$$\boxed{\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}} \quad \text{(*)}$$

$$\frac{(w - 2) + (-2 - w)}{(w - 2) - (-2 - w)} = \frac{(z - 1)(i + 3) + (-1 - z)(-i + 3)}{(z - 1)(i + 3) - (-1 - z)(-i + 3)}$$

$$\frac{w-2-2-w}{w-2+2+w} = \frac{(zi+3z-i-3+i-3+zi-3z)}{(zi+3z-i-3-i+3-zi+3z)}$$

$$\frac{-4}{2w} = \frac{2zi-6}{6z-2i}$$

$$\frac{w}{-2} = \frac{2(3z-i)}{2(zi-3)}$$

$$w = \frac{-2(3z-i)}{(zi-3)}$$

2) Find the Bilinear transform which maps $(1, -i, 0)$ of the z plane into $(0, 2, -i)$ in the w plane.

Soln:

$$\begin{array}{l|l} z_1 = 1 & w_1 = 0 \\ z_2 = -i & w_2 = 2 \\ z_3 = 0 & w_3 = -i \end{array}$$

W.K.T

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-0)(2+i)}{(-2)(-i-w)} = \frac{(z-1)(-i)}{(1+i)(0-z)}$$

$$\frac{w(2+i)}{(-2)(-i-w)} = \frac{(z-1)(-i)}{(1+i)(-z)}$$

$$\frac{w}{(2i+2w)} = \frac{zi-i}{z(1+i)(2+i)}$$

$$\frac{w}{(2i+2w)} = \frac{zi-i}{z(2+i+2i-1)}$$

$$\frac{w}{(2i+2w)} = \frac{zi-i}{z(1+3i)}$$

$$\frac{w}{(2i+2w)} = \frac{zi-i}{z+3iz}$$

$$wz + 3wi z = 2zi^2 - 2i^2 + 2wi z - 2wi$$

$$wz + 3wi z = -2z + 2 + 2wi z - 2wi$$

$$wz + wi z = -2z + 2 - 2wi$$

$$wz + wi z + 2wi = -2z + 2$$

$$w(z+iz+2i) = 2(-z+1)$$

$$w = \frac{2(-z+1)}{(z+iz+2i)}$$

3) Find the Bilinear mapping which maps $z = (1, i, -1)$ into $w = (i, 0, -i)$

Soln:

$$\begin{array}{l|l} z_1 = 1 & w_1 = i \\ z_2 = i & w_2 = 0 \\ z_3 = -1 & w_3 = -i \end{array}$$

W.K.T

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-i)(i)}{(i)(-i-w)} = \frac{(z-1)(i+1)}{(1-i)(-1-z)}$$

$$\frac{(w-i)}{(-i-w)} = \frac{(z-1)(i+1)}{(-1-z)(1-i)} \times \frac{(1+i)}{(1+i)}$$

$$\frac{(w-i)}{(-i-w)} = \frac{(z-1)}{(-1-z)} \frac{(i-1+1+i)}{(1+1)}$$

$$\frac{(w-i)}{(-i-w)} = \frac{(z-1)}{(-1-z)} \frac{(zi)}{2}$$

$$\frac{(w-i)}{(-i-w)} = \frac{zi - i}{-1-z}$$

$$(w-i)(-1-z) = (zi-i)(-i-w)$$

$$-w-wz+i+zi = -zi^2 - ziw + i^2 + iw$$

$$-w-wz+i+zi = z - zi w - 1 + iw$$

$$-w-wz+ziw-iw = z-i-zi-i$$

$$w(-1-z+zi-i) = z-i-zi-i$$

$$w = \frac{z-i-zi-i}{-1-z+zi-i}$$

4) Find the Bilinear transformation which maps

$$z = (0, -i, -1) \text{ into } w = (i, 1, 0)$$

Soln:

$$\begin{array}{l|l} z_1 = 0 & w_1 = i \\ z_2 = -i & w_2 = 1 \\ z_3 = -1 & w_3 = 0 \end{array}$$

W.K.T

$$\frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\frac{(w - i)(1)}{(i - 1)(-w)} = \frac{(z)(-i + 1)}{(i)(-1 - z)}$$

$$\frac{(w - i)}{-w} = \frac{z(-i + 1)(i - 1)}{i(-1 - z)}$$

$$\frac{(w - i)}{-w} = \frac{z(1 + i + i - 1)}{-i - zi}$$

$$\frac{(w - i)}{-w} = \frac{2zi}{-i - zi}$$

$$(w - i)(-i - zi) = 2zi(-w)$$

$$-wi - wzi + i^2 + zi^2 = -2izw$$

$$-wi - wzi - 1 - z + 2izw = 0$$

$$-wi + izw = 1 + z$$

$$w(-i + iz) = z + 1$$

$$\boxed{w = \frac{z + 1}{(-i + iz)}}$$

8)

1) Find the bilinear transform which maps $Z = (0, 1, \infty)$ into the points $w = (i, 1, -i)$

Soln:

Given:

$$\begin{array}{l|l} z_1 = 0 & w_1 = i \\ z_2 = 1 & w_2 = 1 \\ z_3 = \infty & w_3 = -i \end{array}$$

W.K.T

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1) z/z_3 (z_2/z_3 - 1)}{(z_1-z_2) z/z_3 (1 - z/z_3)}$$

$$\frac{(w-i)(1+i)}{(i-1)(-i-w)} = \frac{(z) \left(\frac{1}{\infty} - 1 \right)}{(-1) \left(1 - \frac{z}{\infty} \right)}$$

$$\frac{(w-i)(1+i)}{(i-1)(-i-w)} = \frac{z(-1)}{(-1)(1)}$$

$$\frac{(w-i)}{(-i-w)} = \frac{z(i-1)}{(1+i)} \times \frac{(1-i)}{(1-i)}$$

$$\frac{(w-i)}{(-i-w)} = \frac{z(i+1-1+i)}{1+1}$$

$$\frac{(w-i)}{(-i-w)} = \frac{2iz}{2}$$

$$(w-i) = iz(-i-w)$$

$$w-i = +i^2 z - izw$$

$$w-i = z - izw$$

$$w + izw = z + i$$

$$w(1+iz) = z+i$$

$$w = \frac{z+i}{1+iz}$$

2) Find the bilinear transform for $Z = (0, 1, \infty)$ into the points $w = (1, i, -1)$

Soln:

$$\begin{array}{l|l} z_1 = 0 & w_1 = 1 \\ z_2 = 1 & w_2 = i \\ z_3 = \infty & w_3 = -1 \end{array}$$

W.K.T

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1) \cancel{z_3} (\frac{z_2}{z_3} - 1)}{(z_1-z_2) \cancel{z_3} (1 - \frac{z}{z_3})}$$

$$\frac{(w-1)(i+1)}{(1-i)(-1-w)} = \frac{(z) (\frac{1}{\infty} - 1)}{(-1) (1 - \frac{z}{\infty})}$$

$$[\because \frac{1}{\infty} = 0]$$

$$\frac{(w-1)(i+1)}{(1-i)(-1-w)} = \frac{z(-1)}{(-1)(1)}$$

$$\frac{(w-1)}{(-1-w)} = \frac{z(1-i)}{i+1}$$

$$\frac{(w-1)}{(-1-w)} = \frac{z(1-i)}{(i+1)} \times \frac{(i-1)}{(i-1)}$$

$$\frac{(w-1)}{(-1-w)} = \frac{z(i-1-i^2+i)}{(i^2-1^2)}$$

$$\frac{(w-1)}{(-1-w)} = \frac{z(2i)}{-2}$$

$$(w-1) = -zi(-1-w)$$

$$(w-1) = zi + wzi$$

$$w - wzi = zi + 1$$

$$w(1-zi) = zi + 1$$

$$w = \frac{zi + 1}{1-zi}$$

3. Find the Bilinear Transform which maps the point $z = (\infty, i, 0)$ into $w = (0, -i, \infty)$

soln:

Given:

$$\begin{array}{l|l} z_1 = \infty & w_1 = 0 \\ z_2 = i & w_2 = -i \\ z_3 = 0 & w_3 = \infty \end{array}$$

W.K.T

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-w_1)w_3(\frac{w_2}{w_3}-1)}{(w_1-w_2)w_3(1-\frac{w}{w_3})} = \frac{z_1(\frac{z_2}{z_1}-1)(z_2-z_3)}{z_1(1-\frac{z_2}{z_1})(z_3-z)}$$

$$\frac{(w)(\frac{-i}{\infty}-1)}{(i)(1-\frac{w}{\infty})} = \frac{(\frac{z}{\infty}-1)(i)}{(1-\frac{i}{\infty})(0-z)}$$

$$\frac{w(0-1)}{i(1-0)} = \frac{(0-1)(i)}{(1-0)(-z)}$$

$$\frac{-w}{i} = \frac{-i}{-z}$$

$$-w(-z) = -i \times i$$

$$wz = -i^2$$

$$wz = -(-1)$$

$$\boxed{w = \frac{1}{z}}$$

$$4) z = (0, -1, \infty), w = (-1, -2-i, i)$$

Soln:

Given:

$$\begin{array}{l|l} z_1 = 0 & w_1 = -1 \\ z_2 = -1 & w_2 = -2-i \\ z_3 = \infty & w_3 = i \end{array}$$

W.K.T

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w+1)(-2-i-i)}{(-1+2+i)(i-w)} = \frac{(z)(\frac{-1}{\infty}-1)}{(1)(1-\frac{z}{\infty})}$$

$$\frac{(w+1)(-2-2i)}{(1+i)(i-w)} = \frac{-z}{1}$$

$$\frac{(w+1)}{(i-w)} = -z \frac{(1+i)}{(-2-2i)}$$

$$\frac{(w+1)}{(i-w)} = -z \frac{(1+i)}{(-2-2i)} \times \frac{(-2+2i)}{(-2+2i)}$$

$$\frac{(w+1)}{(i-w)} = -z \frac{(1+i)(-2+2i)}{(-2)^2 - (2i)^2 \cdot 4(-1)}$$

$$\frac{(w+1)}{(i-w)} = -z \frac{(-2 + \cancel{2i} - \cancel{2i} + 2i^2)}{4 + 4}$$

$$\frac{(w+1)}{(i-w)} = -z \frac{(-2-2)}{8}$$

$$\frac{(w+1)}{(i-w)} = -z \frac{(-4)}{8}$$

$$\frac{(w+1)}{(i-w)} = \frac{z}{2}$$

$$2w+2 = zi - zw$$

$$2w + zw = zi - 2$$

$$w(2+z) = zi - 2$$

$$\boxed{w = \frac{zi - 2}{2+z}}$$

Taylor's & Laurent's Series:

Remarks:

Taylor's series:

$$f(z) = f(z_0) + \frac{(z-z_0)}{1!} f'(z_0) + \frac{(z-z_0)^2}{2!} f''(z_0) \dots$$

$$\therefore f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$$

$$\text{where, } a_n = \frac{f^{(n)}(z_0)}{n!}$$

Laurent's series:

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=0}^{\infty} b_n (z-a)^{-n}$$

$$\text{where, } a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-a)^{n+1}} dw$$

$$b_n = \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w-a)^{-n+1}} dw$$

1) Obtain Taylor's series for the function $f(z) = \frac{1}{(z-1)(z-2)}$ in the region $|z| < 1$

Soln:

$$\frac{1}{(z-1)(z-2)} = \frac{A}{(z-1)} + \frac{B}{(z-2)}$$

$$1 = A(z-2) + B(z-1)$$

Put $z=2$

$$\begin{aligned} 1 &= A(2-2) + B(2-1) \\ &= 0 + B(1) \end{aligned}$$

$$1 = B$$

$$\boxed{B=1}$$

Put $z = 1$

$$1 = A(1-2) + B(1-1)$$

$$1 = A(-1) + B(0)$$

$$\frac{1}{-1} = A$$

$$\boxed{A = -1}$$

$$\therefore \Rightarrow \frac{1}{(z-1)(z-2)} = \frac{-1}{(z-1)} + \frac{1}{(z-2)}$$

To find $|z| < 1$:

we have

$$f(z) = \frac{-1}{(z-1)} + \frac{1}{(z-2)}$$

if $|z| < 1$

remove constant term

if $|z| > 1$

remove z term

$$= \frac{-1}{-1 \left[\frac{z}{-1} + 1 \right]} + \frac{1}{-2 \left[\frac{z}{-2} + 1 \right]}$$

$$= \frac{1}{(1-z)} - \frac{1}{2 \left(1 - \frac{z}{2} \right)}$$

$$\left[\because (1-x)^{-1} = 1+x+x^2+x^3+\dots \right]$$

$$(1+x)^{-1} = 1-x+x^2-x^3+\dots$$

$$= [1-z]^{-1} - \frac{1}{2} \left[1 - \frac{z}{2} \right]^{-1}$$

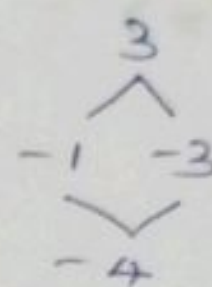
$$= [1+z+z^2+z^3+\dots] - \frac{1}{2} \left[1 + \left(\frac{z}{2} \right) + \left(\frac{z}{2} \right)^2 + \dots \right]$$

$$= \sum_{n=0}^{\infty} z^n - \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2} \right)^n$$

2) Expand the function $f(z) = \frac{1}{z^2 - 4z + 3}$ using Laurent's series valid for the region (i) $|z| > 3$, (ii) $1 < |z| < 3$

Soln:

$$f(z) = \frac{1}{z^2 - 4z + 3}$$



$$f(z) = \frac{1}{(z-1)(z-3)}$$

$$\frac{1}{(z-1)(z-3)} = \frac{A}{(z-1)} + \frac{B}{(z-3)}$$

$$1 = A(z-3) + B(z-1)$$

Put $z = 3$

$$1 = A(3-3) + B(3-1)$$

$$1 = 0 + B(2)$$

$$2B = 1$$

$$\boxed{B = \frac{1}{2}}$$

Put $z = 1$

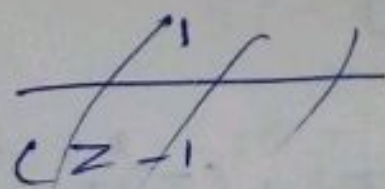
$$1 = A(1-3) + B(1-1)$$

$$1 = A(-2) + 0$$

$$-2A = 1$$

$$\boxed{A = -\frac{1}{2}}$$

$$f(z) = \frac{-\frac{1}{2}}{(z-1)} + \frac{\frac{1}{2}}{(z-3)}$$



i) To find $|z| > 3$

we have

$$f(z) = \frac{-\frac{1}{2}}{(z-1)} + \frac{\frac{1}{2}}{(z-3)}$$

$$= \frac{-\frac{1}{2}}{z[1-\frac{1}{z}]} + \frac{\frac{1}{2}}{z[1-\frac{3}{z}]}$$

$$= \frac{-1}{2z} [1-\frac{1}{z}]^{-1} + \frac{1}{2z} [1-\frac{3}{z}]^{-1}$$

$$= \frac{-1}{2z} [1+(\frac{1}{z})+(\frac{1}{z})^2+\dots] + \frac{1}{2z} [1+(\frac{3}{z})+(\frac{3}{z})^2+\dots]$$

$$= \frac{-1}{2z} \sum_{n=0}^{\infty} (\frac{1}{z})^n + \frac{1}{2z} \sum_{n=0}^{\infty} (\frac{3}{z})^n$$

ii) To find $1 < |z| < 3$

we have

$$f(z) = \frac{-\frac{1}{2}}{(z-1)} + \frac{\frac{1}{2}}{(z-3)}$$

$$\boxed{1 < |z|} \text{ and } \boxed{|z| < 3}$$

no. is checked to the
quest

$$f(z) = \frac{-\frac{1}{2}}{z(1-\frac{1}{z})} + \frac{\frac{1}{2}}{-3(\frac{z}{-3}+1)}$$

$$= \frac{-1}{2z} [1-\frac{1}{z}]^{-1} + -\frac{1}{6} [1-\frac{z}{3}]^{-1}$$

$$= \frac{-1}{2z} [1+(\frac{1}{z})+(\frac{1}{z})^2+\dots] - \frac{1}{6} [1+(\frac{z}{3})+(\frac{z}{3})^2+\dots]$$

i) To find $|z| > 3$

we have

$$f(z) = \frac{-\frac{1}{2}}{(z-1)} + \frac{\frac{1}{2}}{(z-3)}$$

$$= \frac{-\frac{1}{2}}{z[1-\frac{1}{z}]} + \frac{\frac{1}{2}}{z[1-\frac{3}{z}]}$$

$$= \frac{-1}{2z} [1-\frac{1}{z}]^{-1} + \frac{1}{2z} [1-\frac{3}{z}]^{-1}$$

$$= \frac{-1}{2z} [1+(\frac{1}{z})+(\frac{1}{z})^2+\dots] + \frac{1}{2z} [1+(\frac{3}{z})+(\frac{3}{z})^2+\dots]$$

$$= \frac{-1}{2z} \sum_{n=0}^{\infty} (\frac{1}{z})^n + \frac{1}{2z} \sum_{n=0}^{\infty} (\frac{3}{z})^n$$

ii) To find $1 < |z| < 3$

we have

$$f(z) = \frac{-\frac{1}{2}}{(z-1)} + \frac{\frac{1}{2}}{(z-3)}$$

$1 < |z|$ & $|z| < 3$

no. is checked to the
ques

$$f(z) = \frac{-\frac{1}{2}}{z(1-\frac{1}{z})} + \frac{\frac{1}{2}}{-3(\frac{z}{-3}+1)}$$

$$= \frac{-1}{2z} [1-\frac{1}{z}]^{-1} - \frac{1}{6} [1-\frac{z}{3}]^{-1}$$

$$= \frac{-1}{2z} [1+(\frac{1}{z})+(\frac{1}{z})^2+\dots] - \frac{1}{6} [1+(\frac{z}{3})+(\frac{z}{3})^2+\dots]$$

$$= \frac{1}{22} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n - \frac{1}{6} \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$$

3) obtain Taylor's series for the function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ in the region $|z| < 2$, $2 < |z| < 3$

Soln:

$$f(z) = \frac{z^2-1}{(z+2)(z+3)}$$

$$f(z) = \frac{z^2-1}{z^2+5z+6}$$

$$z^2+5z+6 \overline{\left| \begin{array}{r} 1 \\ z^2 \\ -5z-7 \end{array} \right.}$$

$$\frac{5}{4} = A \left(\frac{1}{4} \right)$$

$$\frac{6}{4} = 1 + \frac{2}{4}$$

$$f(z) = 1 + \frac{-5z-7}{(z+2)(z+3)}$$

we have

$$\frac{-5z-7}{(z+2)(z+3)} = \frac{A}{(z+2)} + \frac{B}{(z+3)}$$

$$-5z-7 = A(z+3) + B(z+2)$$

put $z = -2$

$$-5(-2)-7 = A(-2+3) + B(-2+2)$$

$$10-7 = A(1) + 0$$

$$3 = A$$

$$\boxed{A=3}$$

$$= -\frac{1}{22} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n - \frac{1}{6} \sum_{n=0}^{\infty} \left(\frac{z}{3}\right)^n //$$

3) Obtain Taylor's series for the function $f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$ in the region $|z| < 2$, $2 < |z| < 3$

Soln:

$$f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$$

$$f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$$

$$z^2 + 5z + 6 \overline{) \begin{array}{r} 1 \\ z^2 + 5z + 6 \\ \hline -5z - 7 \end{array}}$$

$$\begin{array}{r} 5 = 4 \overline{) 6} \\ \underline{4} \\ 2 \text{ rem } 2 \\ \downarrow \\ \frac{6}{4} = 1 + \frac{2}{4} \\ \text{multiply 4} \end{array}$$

$$f(z) = 1 + \frac{-5z - 7}{(z+2)(z+3)}$$

we have

$$\frac{-5z - 7}{(z+2)(z+3)} = \frac{A}{(z+2)} + \frac{B}{(z+3)}$$

$$-5z - 7 = A(z+3) + B(z+2)$$

Put $z = -2$

$$-5(-2) - 7 = A(-2+3) + B(-2+2)$$

$$10 - 7 = A(1) + 0$$

$$3 = A$$

$$\boxed{A = 3}$$

Put $z = -3$

$$-5(-3) - 7 = A(-3+3) + B(-3+2)$$

$$15 - 7 = 0 + B(-1)$$

$$8 = -B$$

$$B = \frac{8}{-1}$$

$$\boxed{B = -8}$$

$$f(z) = 1 + \frac{3}{(z+2)} - \frac{8}{(z+3)}$$

i) To find $|z| < 2$

$$f(z) = 1 + \frac{3}{2\left[\frac{z}{2} + 1\right]} - \frac{8}{3\left[\frac{z}{3} + 1\right]}$$

$$= 1 + \frac{3}{2} \left[\frac{z}{2} + 1\right]^{-1} - \frac{8}{3} \left[\frac{z}{3} + 1\right]^{-1}$$

$$= 1 + \frac{3}{2} \left[1 - \left(\frac{z}{2}\right) + \left(\frac{z}{2}\right)^2 - \dots\right] - \frac{8}{3} \left[1 - \left(\frac{z}{3}\right) + \left(\frac{z}{3}\right)^2 - \dots\right]$$

If B-series are in alternative put $(-1)^n$ in and

$$f(z) = 1 + \frac{3}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$$

ii) To find $2 < |z| < 3$

we have

$$2 < |z| \text{ \& } |z| < 3$$

$$f(z) = 1 + \frac{3}{(z+2)} - \frac{8}{(z+3)}$$

$$f(z) = 1 + \frac{3}{z(1+\frac{2}{z})} - \frac{8}{3(\frac{z}{3}+1)}$$

$$= 1 + \frac{3}{z} \left[1 + \frac{2}{z} \right]^{-1} - \frac{8}{3} \left[1 + \frac{z}{3} \right]^{-1}$$

$$= 1 + \frac{3}{z} \left[1 - \left(\frac{2}{z} \right) + \left(\frac{2}{z} \right)^2 - \dots \right] - \frac{8}{3} \left[1 - \left(\frac{z}{3} \right) + \left(\frac{z}{3} \right)^2 - \dots \right]$$

$$f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z} \right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3} \right)^n$$

A) Find the Laurent's series for $f(z) = \frac{z^2-4}{(z+1)(z+4)}$
 H.W in region $1 < |z| < 4$

Soln:

$$f(z) = \frac{z^2-4}{(z+1)(z+4)}$$

$$\begin{array}{r} z^2+4z+z+4 \\ z^2+5z+4 \end{array}$$

$$= \frac{z^2-4}{z^2+5z+4}$$

$$\begin{array}{r} 1 \\ z^2+5z+4 \overline{) \begin{array}{r} z^2 \quad -4 \\ z^2+5z+4 \\ \hline -5z-8 \end{array}} \end{array}$$

$$f(z) = 1 + \frac{-5z-8}{(z+1)(z+4)}$$

we have

$$\frac{-5z-8}{(z+1)(z+4)} = \frac{A}{(z+1)} + \frac{B}{(z+4)}$$

$$-5z - 8 = A(z+4) + B(z+1)$$

Put $z = -4$

$$-5(-4) - 8 = A(-4+4) + B(-4+1)$$

$$20 - 8 = 0 + B(-3)$$

$$12 = B(-3)$$

$$B = \frac{12}{-3}$$

$$\boxed{B = -4}$$

Put $z = -1$

$$-5(-1) - 8 = A(-1+4) + B(-1+1)$$

$$5 - 8 = A(3) + 0$$

$$-3 = A(3)$$

$$A = -3/3 \Rightarrow \boxed{A = -1} \quad \therefore f(z) = 1 - \frac{1}{(z+1)} - \frac{4}{(z+4)}$$

To find $1 < |z| < 4$:

$$1 < |z| \text{ \& } |z| < 4$$

$$f(z) = 1 - \frac{1}{z \left[1 + \frac{1}{z} \right]} - \frac{4}{4 \left[\frac{z}{4} + 1 \right]}$$

$$= 1 - \frac{1}{z} \left[1 + \frac{1}{z} \right]^{-1} - \frac{4}{4} \left[1 + \frac{z}{4} \right]^{-1}$$

$$= 1 - \frac{1}{z} \left[1 - \left(\frac{1}{z} \right) + \left(\frac{1}{z} \right)^2 - \dots \right] - (1) \left[1 - \left(\frac{z}{4} \right) + \left(\frac{z}{4} \right)^2 - \dots \right]$$

$$f(z) = 1 - \frac{1}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{z} \right)^n - \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{4} \right)^n$$

5) Find the Laurent's series Expansion for

$$f(z) = \frac{7z-2}{(z+1)z(z-2)} \quad \text{in the region (ii) } |z+1| > 3$$

(i) $1 < |z+1| < 3$

Soln:

Assume $\boxed{z+1=u}$

$$z = u - 1$$

Subs z in $f(z)$

$$f(z) = \frac{7(u-1)-2}{(u-1+1)(u-1)(u-1-2)}$$

$$f(z) = \frac{7u-9}{(u)(u-1)(u-3)}$$

(i) $1 < |u| < 3$ (ii) $|u| > 3$

$$\therefore \frac{7u-9}{u(u-1)(u-3)} = \frac{A}{(u)} + \frac{B}{(u-1)} + \frac{C}{(u-3)}$$

$$7u-9 = A(u-1)(u-3) + B(u)(u-3) + C(u)(u-1)$$

Put $u=1$

$$7(1)-9 = A(1-1)(1-3) + B(1)(1-3) + C(1)(1-1)$$

$$-2 = 0 + B(-2) + C(0)$$

$$-2 = -2B$$

$$B = \frac{-2}{-2} \Rightarrow \boxed{B=1}$$

Put $u = 3$

$$7(3) - 9 = A(3-1)(3-3) + B(3)(3-3) + C(3)(3-1)$$

$$21 - 9 = 0 + 0 + C(3)(2)$$

$$12 = C(6)$$

$$C = \frac{12}{6} \Rightarrow \boxed{C = 2}$$

Put $u = 0$

$$7(0) - 9 = A(0-1)(0-3) + B(0) + C(0)$$

$$-9 = A(-1)(-3)$$

$$-9 = A(3)$$

$$A = -\frac{9}{3} \Rightarrow \boxed{A = -3}$$

$$f(z) = \frac{-3}{u} + \frac{1}{(u-1)} + \frac{2}{(u-3)}$$

i) To find: $1 < |u| < 3$

$$1 < |u| \text{ \& } |u| < 3$$

$$f(z) = -\frac{3}{u} + \frac{1}{u[1-\frac{1}{u}]} + \frac{2}{-3[\frac{u}{-3}+1]}$$

$$= -\frac{3}{u} + \frac{1}{u} \left[1 - \frac{1}{u}\right]^{-1} - \frac{2}{3} \left[1 - \frac{u}{3}\right]^{-1}$$

$$= -\frac{3}{u} + \frac{1}{u} \left[1 + \left(\frac{1}{u}\right) + \left(\frac{1}{u}\right)^2 + \dots \right] - \frac{2}{3} \left[1 + \left(\frac{u}{3}\right) + \left(\frac{u}{3}\right)^2 + \dots \right]$$

$$f(z) = -\frac{3}{u} + \frac{1}{u} \sum_{n=0}^{\infty} \left(\frac{1}{u}\right)^n - \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{u}{3}\right)^n$$

$$f(z) = \frac{-3}{(z+1)} + \frac{1}{(z+1)} \sum_{n=0}^{\infty} \left(\frac{1}{(z+1)}\right)^n - \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{(z+1)}{3}\right)^n$$

ii) To find $|u| > 3$

$$f(z) = \frac{-3}{u} + \frac{1}{u \left[1 - \frac{1}{u}\right]} + \frac{2}{u \left[1 - \frac{3}{u}\right]}$$

$$= -\frac{3}{u} + \frac{1}{u} \left[1 - \frac{1}{u}\right]^{-1} + \frac{2}{u} \left[1 - \frac{3}{u}\right]^{-1}$$

$$= -\frac{3}{u} + \frac{1}{u} \left[1 + \left(\frac{1}{u}\right) + \left(\frac{1}{u}\right)^2 + \dots\right] + \frac{2}{u} \left[1 + \left(\frac{3}{u}\right) + \left(\frac{3}{u}\right)^2 + \dots\right]$$

$$= -\frac{3}{u} + \frac{1}{u} \sum_{n=0}^{\infty} \left(\frac{1}{u}\right)^n + \frac{2}{u} \sum_{n=0}^{\infty} \left(\frac{3}{u}\right)^n$$

$$f(z) = \frac{-3}{(z+1)} + \frac{1}{(z+1)} \sum_{n=0}^{\infty} \left(\frac{1}{(z+1)}\right)^n + \frac{2}{(z+1)} \sum_{n=0}^{\infty} \left(\frac{3}{(z+1)}\right)^n$$

6) Expand using Laurent's series for $f(z) = \frac{z}{(z+1)(z+2)}$

in the region $z = -2$

Soln:

gn:

$$z = -2$$

$$z+2 = u$$

$$\therefore z+2 = u$$

$$z = u - 2$$

$$\therefore f(z) = \frac{u-2}{(u-2+1)(u)}$$

$$f(z) = \frac{u-2}{(u-1)(u)}$$

$$\frac{u-2}{(u-1)(u)} = \frac{A}{(u-1)} + \frac{B}{u}$$

$$u-2 = A(u) + B(u-1)$$

Put $u=0$

$$0-2 = A(0) + B(0-1)$$

$$-2 = 0 + B(-1)$$

$$B = \frac{-2}{-1}$$

$$\boxed{B=2}$$

Put $u=1$

$$1-2 = A(1) + B(1-1)$$

$$-1 = A(1) + 0$$

$$\boxed{A=-1}$$

$$\therefore f(z) = \frac{-1}{(u-1)} + \frac{2}{u}$$

To find $z=-2$:

$$\text{Gm } z=-2$$

If $z = -2$ (or)
some
value
remove const

$\therefore$

$$f(z) = \frac{-1}{-1\left[-\frac{u}{1}+1\right]} + \frac{2}{u}$$

$$= [1-u]^{-1} + \frac{2}{u}$$

$$= [1+u+u^2+\dots] + \frac{2}{u}$$

$$f(z) = \sum_{n=0}^{\infty} (u)^n + \frac{2}{u}$$

$$f(z) = \sum_{n=0}^{\infty} (z+2)^n + \frac{2}{(z+2)}$$

7) Find the Laurent's series for $f(z) = \frac{1}{(z+2)(z^2+1)}$
in the region $1 < |z| < 2$

Soln:

$$f(z) = \frac{1}{(z+2)(z^2+1)}$$

$$\frac{1}{(z+2)(z^2+1)} = \frac{A}{(z+2)} + \frac{Bz+C}{(z^2+1)}$$

$$1 = A(z^2+1) + (Bz+C)(z+2)$$

put $z = -2$

$$1 = A((-2)^2+1) + (B(-2)+C)(-2+2)$$

$$1 = A(4+1) + 0$$

$$1 = A(5)$$

$$\boxed{A = \frac{1}{5}}$$

$$z^2+1=0$$

$$z^2 = -1$$

$$z = \pm i$$

11/11/2022

Transformation problem:

1) The transformation $w = \sin z$

Soln

$$w = \sin z$$

$$u + iv = \sin(x + iy)$$

$$[\sin(A+B) = \sin A \cos B + \cos A \sin B]$$

$$[\therefore \sin(i0) = i \sinh 0$$

$$\cos(i0) = \cosh 0]$$

$$u + iv = \sin x \cos(iy) + \cos x \sin(iy)$$

$$= \sin x \cosh y + i \cos x \sinh y$$

Separate real & img

$$u = \sin x \cosh y \quad \text{--- (1)}$$

$$v = \cos x \sinh y \quad \text{--- (2)}$$

case(i)

for removing we put const value

$x = a$ (||^{le} to y axis) in eqn (1) & (2)
to eliminate y

case(ii) $y = b$

to eliminate x

$$u = \sin a \cosh y$$

$$v = \cos a \sinh y$$

$$\frac{u}{\sin a} = \cosh y$$

$$\frac{v}{\cos a} = \sinh y$$

Squaring & subtracting (because we need 1 to sub the term)

$$\left(\frac{u^2}{\sin^2 a} \right) - \left(\frac{v^2}{\cos^2 a} \right) = \cosh^2 y - \sinh^2 y$$

(if u & v were
we sub the
term to get
(1))

$$\frac{u^2}{\sin^2 a} - \frac{v^2}{\cos^2 a} = 1$$

$$\text{with } \frac{u^2}{A^2} - \frac{v^2}{B^2} = 1$$

$$\text{where } A = \sin a, b = \cos a$$

The above eqn represent hyperbola

case(ii)

$y = b$ (||^{le} to x axis) in eqn ① & ②

$$\begin{array}{l|l} u = \sin x \cosh b & v = \sin x \cos x \sinh b \\ \frac{u}{\cosh b} = \sin x & \frac{v}{\sinh b} = \cos x \end{array}$$

squaring & ~~subtract~~ Adding

$$\left(\frac{u^2}{\cosh^2 b} \right) + \left(\frac{v^2}{\sinh^2 b} \right) = \sin^2 x + \cos^2 x$$

$$\left(\frac{u^2}{\cosh^2 b} \right) + \left(\frac{v^2}{\sinh^2 b} \right) = 1$$

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

$$\text{where } A = \cosh b, b = \sinh b$$

The above eqn represent ellipse

2) The transformation $w = \cos z$

Soln:

Given :

$$w = \cos z$$

$$u + iv = \cos(x + iy)$$

$$\therefore \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\begin{aligned} \therefore \cos(i0) &= \cosh 0 \\ \sin(i0) &= i \sinh 0 \end{aligned}$$

$$u + iv = \cos x \cos iy - \sin x \sin iy$$

$$= \cos x \cosh y - i \sin x \sinh y$$

$$u = \cos x \cosh y \quad \text{--- ①}$$

$$v = -\sin x \sinh y \quad \text{--- ②}$$

Case (i) $x = a$ (||^{le} to y axis) in eqn ① & ②

$$u = \cos a \cosh y$$

$$v = -\sin a \sinh y$$

$$\frac{u}{\cos a} = \cosh y$$

$$\frac{v}{-\sin a} = \sinh y$$

squaring & subtracting

$$\frac{u^2}{\cos^2 a} - \frac{v^2}{-(\sin^2 a)} = \cosh^2 y - \sinh^2 y$$

$$\frac{u^2}{\cos^2 a} - \frac{v^2}{\sin^2 a} = 1$$

$$\frac{u^2}{A^2} - \frac{v^2}{B^2} = 1$$

where ,

$$A = \cos^2 a, \quad B = \sin^2 a$$

The above eqn represent ~~ellipse~~ ^{hyperbola}

Case (ii) :

$y = b$ (||^{le} to x axis) in eqn ① & ②

$$u = \cos x \cosh b$$

$$v = -\sin x \sinh b$$

$$\frac{u}{\cosh b} = \cos x$$

$$\frac{v}{-\sinh b} = \sin x$$

Squaring & Adding

$$\frac{u^2}{\cos^2 hb} + \frac{v^2}{(-\sin^2 hb)} = \cos^2 x + \sin^2 x$$

$$\frac{u^2}{\cos^2 hb} + \frac{v^2}{\sin^2 hb} = 1$$

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

where,

$$A = \cosh b, \sinh b = B$$

The above eqn represent ^{ellipse} hyperbola.

Image Problem:

1) Find the image of circle $|z-1|=1$ under the mapping

$$w = \frac{1}{z}$$

Soln:

Given:

$$|z-1|=1$$

$$\text{where, } z = x+iy$$

$$|x+iy-1|=1$$

$$|(x-1)+iy|=1$$

$$\sqrt{(x-1)^2 + y^2} = 1$$

squaring on both sides

$$(x-1)^2 + y^2 = 1$$

$$x^2 + 1 - 2x + y^2 = 1$$

$$x^2 + y^2 - 2x = 0 \quad \text{--- (1)}$$

$$w = \frac{1}{z}$$

$$u+iv = \frac{1}{x+iy}$$

$$x+iy = \frac{1}{u+iv}$$

$$x+iy = \frac{1}{u+iv} \times \frac{u-iv}{u-iv}$$

$$x+iy = \frac{u-iv}{u^2+v^2}$$

$$x = \frac{u}{u^2+v^2} \quad \Bigg| \quad y = \frac{-v}{u^2+v^2}$$

Subs in eqn ①

$$\frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} - 2\left(\frac{u}{u^2+v^2}\right) = 0$$

$$\frac{u^2+v^2-2u(u^2+v^2)}{(u^2+v^2)^2} = 0$$

$$(u^2+v^2)-2u(u^2+v^2) = 0$$

$$(u^2+v^2)(1-2u) = 0$$

$$1 \times 0 = 0$$

$$\boxed{1-2u=0}$$

It is straight line

$$\boxed{x^2+y^2=a^2} \begin{matrix} \nearrow \text{radius} \\ \rightarrow \text{circle} \end{matrix}$$

circle is not equal to 0. because without radius we can't draw a circle

2) Find the image $|z+2i|=2$ under the mapping

$$w = \frac{1}{z}$$

Soln:

Given:

$$|z+2i|=2$$

$$|x+iy+2i|=2$$

$$|x+i(y+2)|=2$$

$$\sqrt{x^2+(y+2)^2}=2$$

Squaring on both sides

$$x^2+(y+2)^2=4$$

$$x^2+y^2+\cancel{4}+4y=\cancel{4}$$

$$x^2+y^2+4y=0 \quad \text{--- (1)}$$

$$w = \frac{1}{z}$$

$$u+iv = \frac{1}{x+iy}$$

$$x+iy = \frac{1}{u+iv}$$

$$= \frac{1}{u+iv} \times \frac{u-iv}{u-iv}$$

$$x+iy = \frac{u-iv}{u^2+v^2}$$

$$x = \frac{u}{u^2+v^2} \quad \Bigg| \quad y = \frac{-v}{u^2+v^2}$$

Subs in eqn ①

$$\frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} + 4 \left(\frac{-v}{(u^2+v^2)} \right) = 0$$

$$\frac{u^2 + v^2 - 4v(u^2 + v^2)}{(u^2 + v^2)^2} = 0$$

$$u^2 + v^2 - 4v(u^2 + v^2) = 0$$

$$(u^2 + v^2)[1 - 4v] = 0$$

$$\boxed{1 - 4v = 0}$$

$$x^2 + y^2 = a^2$$

It is straight line.

3) Find the image of $|z| = 2$ under the transformation

$$w = 3z$$

Soln:

Given:

$$|z| = 2$$

$$|x + iy| = 2$$

$$\sqrt{x^2 + y^2} = 2$$

squaring on both sides

$$x^2 + y^2 = 4$$

$$x^2 + y^2 - 4 = 0 \quad \text{--- ①}$$

$$w = 3z$$

$$u + iv = 3(x + iy)$$

$$x + iy = \frac{u + iv}{3}$$

$$x = \frac{u}{3}, \quad y = \frac{v}{3}$$

$$\frac{(u)^2}{(3)^2} + \frac{v^2}{3^2} - 4 = 0$$

$$\frac{u^2 + v^2 - 36}{9} = 0$$

$$u^2 + v^2 - 36 = 0$$

$$u^2 + v^2 = 36$$

$$\boxed{u^2 + v^2 = (6)^2}$$

It is circle with radius 6

2 marks:

1) Define conformal mapping:

The mapping that preserves angles between any oriented curves both in magnitude and direction is called conformal mapping.

2) Define critical point:

The point in which $f'(z) = 0$ is called critical point. At this point transformation is not conformal.

3) Define cross ratio of Bilinear transformation

If z_1, z_2, z_3, z_4 are complex numbers then

$$\frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}$$

4) Define Singular point

If $f(z)$ fails to be analytic at a point $z = a$ is called singular point

5) Define Essential singularity

If the principle part contains infinite number of terms with negative power of $(z-a)$ then $z=a$ is known as essential singularity.

Example: $f(z) = e^{1/z}$

$$= 1 + \frac{1}{\frac{1}{z}} + \frac{\left(\frac{1}{z}\right)^2}{2!} + \dots$$

1) Discuss the transformation $w = \cosh z$.

Soln:

Given:

$$w = \cosh z$$

$$w = \cos iz$$

$$[\because \cosh 0 = \cos i0]$$

$$i \sinh 0 = \sin i0$$

$$u + iv = \cos i(x + iy)$$

$$= \cos(x - y)$$

$$[\because \cos(A - B) = \cos A \cos B + \sin A \sin B]$$

$$u + iv = \cos x \cos y + \sin x \sin y$$

$$= \cosh x \cos y + i \sinh x \sin y$$

$$u = \cosh x \cos y \text{ --- (1), } v = \sinh x \sin y \text{ --- (2)}$$

case (i) $x = a$ (Parallel to y-axis) in eqn (1) & (2)

$$u = \cosh a \cos y, \quad v = \sinh a \sin y$$

$$\frac{u}{\cosh a} = \cos y, \quad \frac{v}{\sinh a} = \sin y$$

squaring & Adding

$$\frac{u^2}{\cosh^2 a} + \frac{v^2}{\sinh^2 a} = \cos^2 y + \sin^2 y$$

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

$\therefore$ The above eqn represent ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

where,

$$A = \cosh a$$

$$B = \sinh a$$

case (ii) $y = b$ (Parallel to x-axis) in eqn (1) & (2)

$$u = \cosh x \cos b, \quad v = \sinh x \sin b$$

$$\frac{u}{\cos b} = \cosh x, \quad \frac{v}{\sin b} = \sinh x$$

squaring & Subtracting

$$\frac{u^2}{\cos^2 b} - \frac{v^2}{\sin^2 b} = \cosh^2 x - \sinh^2 x$$

$$\frac{u^2}{A^2} - \frac{v^2}{B^2} = 1$$

The above eqn represent on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

where,

$$A = \cos b$$

$$B = \sin b$$

2) Discuss the transformation $w = \sinh z$

Soln:

Given:

$$w = \sinh z$$

$$w = \frac{1}{i} [\sin i(x+iy)]$$

$$\begin{aligned} [\because \sin(i\theta) &= i \sinh \theta \\ \sinh \theta &= \frac{1}{i} \sin i\theta] \end{aligned}$$

$$u+iv = \frac{1}{i} [\sin(ix-y)]$$

$$= \frac{1}{i} [\sin ix \cos y - \cos ix \sin y]$$

$$= \frac{1}{i} [i \sinh x \cos y - \cosh x \sin y]$$

$$= \sinh x \cos y + i \cosh x \sin y \quad [\because \frac{1}{i} = -i]$$

$$u = \sinh x \cos y, \quad v = \cosh x \sin y \quad \text{--- (2)}$$

Case (i) $x = a$ ($\parallel$ to y -axis) in eqn ① & ②

$$u = \sinh a \cos y, \quad v = \cosh a \sin y$$

$$\frac{u}{\cosh a} = \cos y, \quad \frac{v}{\sinh a} = \sin y$$

squaring & adding

$$\frac{u^2}{\sinh^2 a} + \frac{v^2}{\cosh^2 a} = \cos^2 y + \sin^2 y$$

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

The above eqn represent an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

where;

$$A = \sinh a$$

$$B = \cosh a$$

Case (ii) $y = b$ (||^l to x -axis) in eqn ① & ②

$$u = \sinh x \cos b, \quad v = \cosh x \sin b$$

$$\frac{u}{\cos b} = \sinh x, \quad \frac{v}{\sin b} = \cosh x$$

squaring & subtracting

$$\frac{u^2}{\cos^2 b} - \frac{v^2}{\sin^2 b} = \sinh^2 x \cosh^2 x - \cosh^2 x \sinh^2 x$$

$$\frac{v^2}{A^2} - \frac{u^2}{B^2} = 1$$

The above eqn represent on hyperbola $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

where,

$$A = \sin b$$

$$B = \cos b$$